WARD: A Vision-Only Framework for Water-Surface Awareness and Risk Detection

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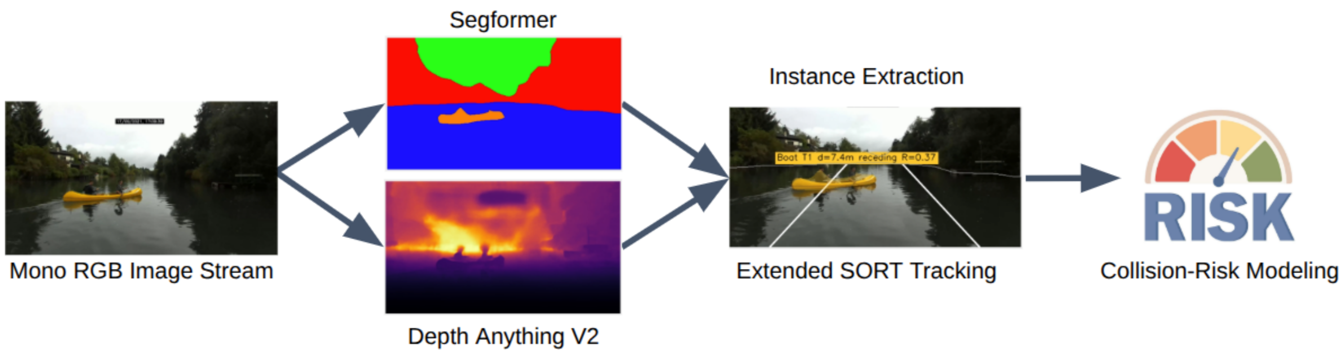

Fig. 1: Overview of the proposed framework. Monocular RGB frames are processed by SegFormer and Depth Anything V2 to extract semantic and depth cues, which are then fused to generate actionable safety warnings.

***Abstract*—Safe navigation for Autonomous Surface Vehicles (ASVs) requires both accurate obstacle detection and actionable risk assessment. We propose WARD (Water-surface Awareness and Risk Detection), a vision-only, risk-aware framework that can output safety assessments for low-cost ASVs. WARD fuses SegFormer semantic segmentation with radar-calibrated Depth Anything V2 monocular depth using a novel continuous collision-corridor risk model. The risk model reasons jointly about proximity, lateral offset, and semantic hazards to produce human-interpretable safety alerts. An extended SORT tracker ensures stable Time-to-Collision predictions by filtering monocular depth noise. Operating at 13.7 FPS on a consumer-grade GPU, WARD provides reliable, discrete safety warnings (SAFE, CAUTION, IMMEDIATE) for resource-constrained ASVs.**

I. Introduction

Autonomous Surface Vehicles (ASVs) are increasingly deployed in unstructured maritime environments such as busy ports and marinas. These domains present unique perception challenges, including dynamic water reflections, waves, and low-profile hazards. While high-end platforms rely on active sensors like marine radar or LiDAR, small-scale ASVs demand low-cost, vision-only solutions that operate reliably under strict power and hardware constraints.

In safety-critical navigation, a gap exists between perception accuracy and operational safety: high accuracy in perception modules, such as object detection and segmentation, does not inherently translate into safe navigation. The fundamental question for an ASV is not whether an obstacle is perfectly segmented, but a more urgent one: *"Does this obstacle pose an immediate collision risk?"* A vision system that returns raw detections without this risk-aware context can overwhelm the controller with data while failing to highlight the most pressing threats, such as treating a distant, harmless buoy and a swimmer in the immediate path as equally salient.

To bridge this gap, we propose WARD (Water-surface Awareness and Risk Detection), a vision-only (RGB), risk-aware framework that fuses SegFormer [1] semantic segmentation with radar-calibrated Depth Anything V2 [2] monocular depth, as illustrated in Fig. 1. WARD introduces a continuous collision-corridor risk model to reason jointly about proximity, lateral offset, closing velocity, and semantic hazard. By delivering actionable, real-time safety assessments (SAFE, CAUTION, and IMMEDIATE) from a single RGB stream, WARD offers a practical baseline for resource-constrained ASVs operating in dynamic environments.

II. Related Work

A. Maritime Semantic Segmentation

Fully-convolutional architectures such as U-Net [3] and the DeepLab family [4] established atrous convolution and spatial-pyramid pooling as standard tools for enlarging the effective receptive field, and transformer-based SegFormer [1] later matched or exceeded these CNN baselines with a lighter hierarchical-attention design. In the maritime domain, WaSR [5] and MODD [6] adapted this line of work to cope with wakes, reflections, and sun glitter, and the LaRS [7] and WaterScenes [8] datasets have since consolidated training and evaluation data for ASV perception, with LaRS serving as our primary source. Nearly all of this work, however, optimizes pixel-level mIoU as the end goal, a metric that rewards precise boundaries but is silent on whether a hazard was detected at all. We instead treat segmentation as one stage of a downstream warning pipeline: evaluation is re-scaled around obstacle recall and inference latency, and SegFormer is extended with a boundary-prediction head, absent from the original design, to recover instance separation from a semantic-only backbone.

B. Monocular Depth Estimation

Monocular depth estimation has moved from dataset-specific regressors toward generalist foundation models. MiDaS [9] showed that mixing depth datasets during training enables zero-shot relative-depth transfer, Depth Anything [10] scaled this idea to tens of millions of unlabeled training images, and Depth Anything V2 [2] sharpened object boundaries with synthetic data and a larger teacher network. Diffusion-based methods such as Marigold [11] and DepthFM [12] trade inference speed for finer detail recovery. All of these models are trained primarily on road-driving and indoor imagery [13]; applied directly to open water, they show a consistent domain gap, with the horizon saturating and metric scale drifting away from physical units. Rather than retraining on scarce maritime depth data, we keep the pre-trained Depth Anything V2 representation intact and correct this drift with a single global scaling factor fit against radar ground truth, restoring near-range metric accuracy in the close-up range that matters most for collision warning.

C. Multi-Object Tracking

SORT [14] is a widely-used online tracker that combines a constant-velocity Kalman filter on 2D bounding boxes with Hungarian matching on IoU. DeepSORT [15] augments SORT with a learned appearance embedding to reduce identity switches through occlusion. We adopt the SORT formulation for its low latency and minimal training requirements, but extend its 7-dimensional Kalman state to nine dimensions so that metric depth and its rate of change are tracked jointly with bounding-box dynamics.

D. Collision-Risk Modeling

Two lines of thought have shaped this area. The first is formal safety, exemplified by Mobileye's Responsibility-Sensitive Safety (RSS) framework [16], which derives longitudinal and lateral safe distances from reaction time and braking capability, and by the ISO 22839 standard for forward collision mitigation [17]. The second relies on domain-specific heuristics: the ship-domain concept of Fujii and Tanaka [18] sizes a safe region around a vessel from its beam and operating speed, and recent ASV work adapts COLREGs rules to reactive collision avoidance [19]. Both typically collapse risk to a binary safe/unsafe decision and assume range measurements from active sensors. Our model keeps RSS's geometric decomposition into a forward-range term and a lateral-excess term, but replaces the kinematic safe-distance equations with a continuous exponential-decay score, weighted by obstacle class and modulated by closing velocity, so that approaching and receding obstacles at the same range are no longer treated as equally risky.

III. Technical Approach

WARD's modular pipeline processes RGB frames through parallel SegFormer and Depth Anything V2 modules. These outputs are fused via our continuous collision-corridor risk model to generate per-obstacle risk scores. To ensure stability, an extended SORT tracker smooths these scores before mapping to discrete warning levels.

A. Maritime Semantic Segmentation

We adopt SegFormer (MiT-B0) trained on the LaRS [7] dataset as our segmentation backbone. To enable instance separation from a semantic-only backbone, we extend it with a dual-head design: a 9-class semantic head and an auxiliary obstacle-boundary head, the latter used to split adjacent same-class instances during connected-component extraction.

LaRS provides both semantic and instance-level annotations under conditions such as water reflections, sun glitter, and partially submerged obstacles. The MiT-B0 encoder was selected among the architectures we benchmarked for its accuracy–latency trade-off. The nine semantic classes, Static Obstacle, Water, Sky, Boat, Buoy, Swimmer, Animal, Float, and Other, are obtained by merging the LaRS panoptic categories. The two heads are trained jointly with a combined cross-entropy, Dice, and boundary binary cross-entropy loss, with class-specific weights (e.g., $5\times$ for Swimmer, $4\times$ for Buoy) to counter the severe class imbalance among hazard categories. At inference, the segmentation mask and calibrated depth map are fused: boundary pixels above a threshold $\tau_b = 0.4$ are first suppressed to split adjacent same-class instances, and connected-component analysis is then applied to every non-background class (excluding Water and Sky) to isolate contiguous obstacle regions, discarding regions smaller than 300 px as segmentation noise. Each surviving instance is assigned a representative depth equal to the 5th percentile of its pixel-wise depth distribution, a conservative estimate of its nearest point, and a normalized lateral offset $\ell \in [-1, 1]$ relative to the image center.

B. Monocular Depth Estimation

Depth is estimated using the Metric-Outdoor variant of Depth Anything V2, fine-tuned from the foundation model on synthetic Virtual KITTI data [13]. To correct scale bias from road-centric pre-training without retraining on scarce maritime data, we apply a single global scaling factor $s_{cal} = 0.4$ calibrated once against radar ground truth from the WaterScenes dataset [8] according to $d_{\text{cal}} = s_{cal} \cdot d_{\text{raw}}$, where d_{raw} is the raw output from the Depth Anything V2 model and d_{cal} represents the calibrated metric depth in meters. This adjustment restores near-metric accuracy in the critical 0–20 m close-up range, which is the primary zone for maritime collision warning.

The scaling factor is fit using the sample subset of WaterScenes, which provides synchronized RGB and 4D-radar streams; radar points are projected onto image coordinates via the dataset's calibration to yield per-pixel metric ground truth. Even after calibration, accuracy degrades beyond 20 m, and frame-to-frame estimates retain residual noise (median inter-frame variation of 0.28 m on our evaluation sequences). Instead of pursuing tighter calibration, we design the downstream modules to tolerate this noise: the risk model relies on a conservative per-instance depth percentile rather than raw pixel depth, the tracker below smooths depth over time, and warning-level hysteresis absorbs the remaining flicker.

C. Multi-Object Tracking

To mitigate monocular depth noise, we extend the SORT tracker to a 9-D Kalman state $\mathbf{x} = [c_x, c_y, s, r, d, \dot{c}_x, \dot{c}_y, \dot{s}, \dot{d}]^\top$, where (c_x, c_y), s, and r denote the bounding-box center, scale (area), and aspect ratio; d is the calibrated metric depth; and dotted variables represent their respective rates of change. The filter optimally balances depth observation noise against a constant-velocity process model. This yields smoothed estimates of the closing velocity, $v_{\text{closing}} = -\dot{d} \cdot f_{\text{fps}}$, and the Time-to-Collision, $TTC = d/v_{\text{closing}}$, where f_{fps} is the system frame rate. These parameters feed directly into the risk model in Eq. (2) without requiring post-hoc exponential moving averages or windowing.

Data association uses the Hungarian algorithm on an IoU cost matrix between predicted and observed boxes, with a minimum IoU of 0.20; unmatched detections spawn new tracks, tracks left unmatched for five consecutive frames are deleted, and a track is confirmed after two consecutive hits. Within that five-frame grace window, the filter continues to propagate depth and bounding-box motion forward under the constant-velocity model even without a matching detection, so a brief occlusion degrades the closing-velocity estimate gracefully rather than resetting it outright. The normalized lateral offset of each track is separately smoothed with an exponential moving average ($\alpha = 0.4$), which keeps small shifts in the segmentation mask from producing abrupt changes in the lateral term of the risk model below.

D. Collision-Risk Modeling

Based on the Responsibility-Sensitive Safety (RSS) framework [16], we define a virtual forward collision corridor of width $W_{\text{boat}} + 2M_{\text{lat}}$ centered on the ASV heading, where W_{boat} is the ASV beam width and M_{lat} is the lateral safety margin. For an obstacle with forward distance d_{fwd} and lateral excess e_{lat} outside this corridor, we first compute a static risk baseline:

$$R_{\text{base}} = w_{\text{class}} \cdot \exp(-d_{\text{fwd}}/D_{\text{safe}}) \cdot \exp(-e_{\text{lat}}/L_{\text{safe}}) \,, \quad (1)$$

where w_{class} is the semantic hazard weight, D_{safe} is the forward decay constant, and L_{safe} is the lateral decay constant. This baseline is amplified by a closing-velocity factor:

$$R = R_{\text{base}} \cdot \big(1 + \alpha_v \cdot \min(v_{\text{closing}}/V_{\text{ref}},\ 2)\big), \qquad (2)$$

where v_{closing} is the Kalman-filtered closing velocity, V_{ref} is the reference speed, and α_v is the amplification factor. The continuous score R is mapped to three discrete warning levels, SAFE ($R < 0.15$), CAUTION ($0.15 \le R < 0.50$), and IMMEDIATE($R \ge 0.50$), via threshold hysteresis with separate enter/exit values to suppress noise-induced flicker.

Concretely, $W_{\text{boat}} = 2\,\text{m}$ and $M_{\text{lat}} = 1\,\text{m}$, giving the lateral excess $e_{\text{lat}} = \max(0,\ |d_{\text{lat}}| - (W_{\text{boat}}/2 + M_{\text{lat}}))$, where $d_{\text{lat}} = \ell \cdot d_{\text{fwd}} \cdot \tan(\theta_{\text{HFOV}}/2)$. The decay constants are $D_{\text{safe}} = 12\,\text{m}$ and $L_{\text{safe}} = 2\,\text{m}$; the per-class weight w_{class} runs from 1.0 for Boat down to 0.7 for Buoy and 0.3 for Static Obstacle; and the velocity term uses $V_{\text{ref}} = 3\,\text{m/s}$ with $\alpha_v = 1.0$. The multiplicative form requires both proximity and lateral alignment to be threatening at once, while the velocity term penalizes an approaching obstacle more heavily than a stationary or receding one at the same range. An obstacle directly ahead at $d_{\text{fwd}} = 0$ returns $R = w_{\text{class}}$, and with the velocity factor included the score can reach $w_{\text{class}} \cdot (1 + 2\alpha_v)$, so a high-weight obstacle closing rapidly can exceed 1.0 rather than saturate at a nominal ceiling. In practice, the enter/exit hysteresis thresholds for IMMEDIATE are set at 0.50 and 0.38, respectively, so a warning only drops once the score falls comfortably below the value that triggered it.

IV. Results

V. Conclusion

We presented WARD, a lightweight, vision-only framework that bridges raw perception and actionable risk assessment. By fusing dual-head SegFormer segmentation with radar-calibrated Depth Anything V2 depth, WARD recovers near-range metric accuracy without the cost and complexity of active ranging hardware, while the extended SORT tracker and collision-corridor risk model turn noisy per-frame estimates into stable, human-interpretable warnings. Operating at real-time rates on a consumer-grade GPU, WARD enables safe navigation for low-cost ASVs.

Future work will pursue field deployment on a physical ASV to test robustness against vessel heave and pitch, extend the architecture to a multi-camera configuration for 360° situational awareness through cross-camera tracking, and close the loop by feeding the continuous risk scores into path-planning and control for autonomous evasive maneuvers. A synthetic evaluation set built with generative video models would also help offset the scarcity of annotated maritime hazard data.

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